

[Secure Exit](#)SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT1-Assignment

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QUESTIONS

Q1Banking Transaction System –
Exception Handling

10 marks

**Q2**Hospital Patient Registration –
User-Defined Exception

10 marks

**Q3**E-Commerce Order Processing
– Built-in Exceptions

10 marks

**Q4**Online Food Delivery – Multiple
Threads and Execution Order

10 marks

**Q5**Manufacturing Plant –
Producer-Consumer Problem

10 marks



5/5 answered

Q1

Banking Transaction System – Exception Handling

10
marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario	Expected Result	Purpose
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10000 3000	Normal case	
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10000 12000	Boundary / exception case	
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10000 -500	Invalid input	
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abc 3000	Invalid input	
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Your Code

```
1 import java.util.*;
2 public class InsufficientbalanceException ex
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         try
8         {
9             double balance = sc.nextDouble();
10            double withdraw = sc.nextDouble();
11            if(balance<0||withdraw<=0)
12            {
13                throw new IllegalArgumentExc
14            }
15            if(withdraw>balance)
16            {
17                throw new Insufficientbalanc
18            }
19            double remaining = balance-withd
20            System.out.println("successful")
21            System.out.println(withdraw);
22            System.out.println(remaining);
23        }
24        catch(InsufficientbalanceException e
25        {
26            System.out.println("insufficient
27        }
28        catch(Exception e)
29        {
30            System.out.println("invalid input
31        }
32        finally
33        {
34            System.out.println("completed");
35        }
36        sc.close();
37    }
38 }
39 }
40 }
```

Input

abc
3000



Output

```
invalid input  
completed
```

✔ Submitted

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